ResearchAgent



Paid on delivery.
Proven on the ledger.

ResearchAgent is research with a wallet. It answers from free sources first. Policy can buy an article within your budget; the writer collects by publishing its key on XRPL Testnet.

Ask a question	https://ra-front-651013803640.asia-southeast1.run.app · Shared XRPL Testnet demo · live answers, rule-based buying · no real value
Recorded run	collaboration95.github.io/theFastandtheFungible/demo/ · a recorded Testnet run, any time

Check the recorded receipts in the judges' corner — no question used.

```
$ npx -y https://github.com/Collaboration95/theFastandtheFungible/releases/download/v1.1.1/researchagent-mcp-1.1.1.tgz try --testnet
```

Since SingHacks: delivery, identity and receipts you can check

At SingHacks a language model proposed each purchase and a person approved every quote before payment; payment was a fixture by default, with optional Testnet settlement.

	At SingHacks	Now
Who decides to buy	A language model proposed; a person approved every quote	Policy code buys within your budget after one plan confirmation; a decision model (fixed rules in the sandbox) only scores
Fair exchange	Pay first, then read	Key-reveal escrow: the writer is paid only by publishing the article's key
Refunds	None	A broken promise can receive a writer-signed refund; a real writer could refuse
Proof	Citations checked inside the app	A browser key check and recorded-run verifier; TypeScript and Python tools for exported receipts
Identity	No operator credential domain	Three accepted Testnet writer Credentials; one Testnet Permissioned Domain for identity display

304
pull requests merged since 21 Sep

15
releases, v0.1.0 to v1.1.1

61
Testnet transactions you can re-check live on our ledger page

224
receipt test vectors; two verifiers, written separately, agree on all

Counted at 88c3a00b on 12 Nov 2026 (git and gh queries in site/progress/data.json) · "At SingHacks" is the 19–20 Sep event build (e16435a3)

One line runs the whole money path on real Testnet

```
$ npx -y https://github.com/Collaboration95/theFastandtheFungible/releases/download/v1.1.1/researchagent-mcp-1.1.1.tgz try --testnet
```

```
Testnet · no real value · Fixture models · fictional and team-authored writers
1. Create: 4AF8D55C…452022B6
2. Finish · sha256(K) = Condition ✓: 47FB0F5C…E9E83FA6
3. Refund · writer-signed · fixture writer cooperates (a real writer could refuse): 6A449846…702CE755
4. Re-buy create: DF565032…4E4EF373
5. Re-buy finish · sha256(K) = Condition ✓: F1AA7E20…5DB3A95F
6. Anchor: A722EB98…6FACB8AF
VERIFY 7cb2a7a4-2f7b-40d0-a84d-f5dbbf65fb98
PASS CONTRIBUTIONS 1/1
PASS MANIFEST   7/7
PASS PASSAGES   7/7
PASS CITATION   20/20
PASS GRANT      1/1
PASS RECEIPT    3/3
PASS REFUND     2/2
PASS AUDIT CHAIN 1/1
PASS testnet 12/12 (read-only, s.altnet)
Skipped writer credentials: the writer's identity is self-declared; checking it needs an online lookup
Skipped historical origin: not needed for this receipt format (one origin per article)
PASS audit anchor: Validated XRPL Testnet ledger 22238845
Skipped is not failed. PASS does not prove who a writer is.
Finished in 59.2 s · budget S$2.00 · net S$0.40 · gross S$0.70
RESULT PASS
```

84.2 s

from an empty npm cache to six Testnet links and RESULT PASS: buy, refund a broken promise, re-buy, anchor, verify. One cold run on one Mac.

2 of 4

cold runs overran the 90 s target. We keep the misses.

This is a recorded v1.1.1 run; its clock excludes the npm download. A faucet fallback is labelled SIMULATED. Later source features are not inferred to be in this release.

— What we built

One question, one budget, and only code can spend it

1
Ask
A question and a budget you set.

2
Confirm the plan
A short sheet confirms the plan, not a purchase.

3
Search writers
Writers return abstracts and signed promises, never premium text.

4
Decide
A decision model (fixed rules in the live sandbox) scores what is worth buying, weighted by trust.

5
Pay
Policy checks the budget, dry-runs the unsigned create, then signs and submits on Testnet.

6
Verify and answer
Decrypt, check signed promises, request a writer-signed refund if broken, cite, anchor.

Five hard gates, on every run

1. No premium bytes before a grant.
2. The budget is the only spending authority; only policy code can buy, never the language model.
3. One charge per intent, across retries and restarts.
4. Every citation points at a delivered passage.
5. Anything simulated or substituted is labelled.

0 gate violations

across the v5 held-out and off-topic question banks on Testnet (11 Nov).

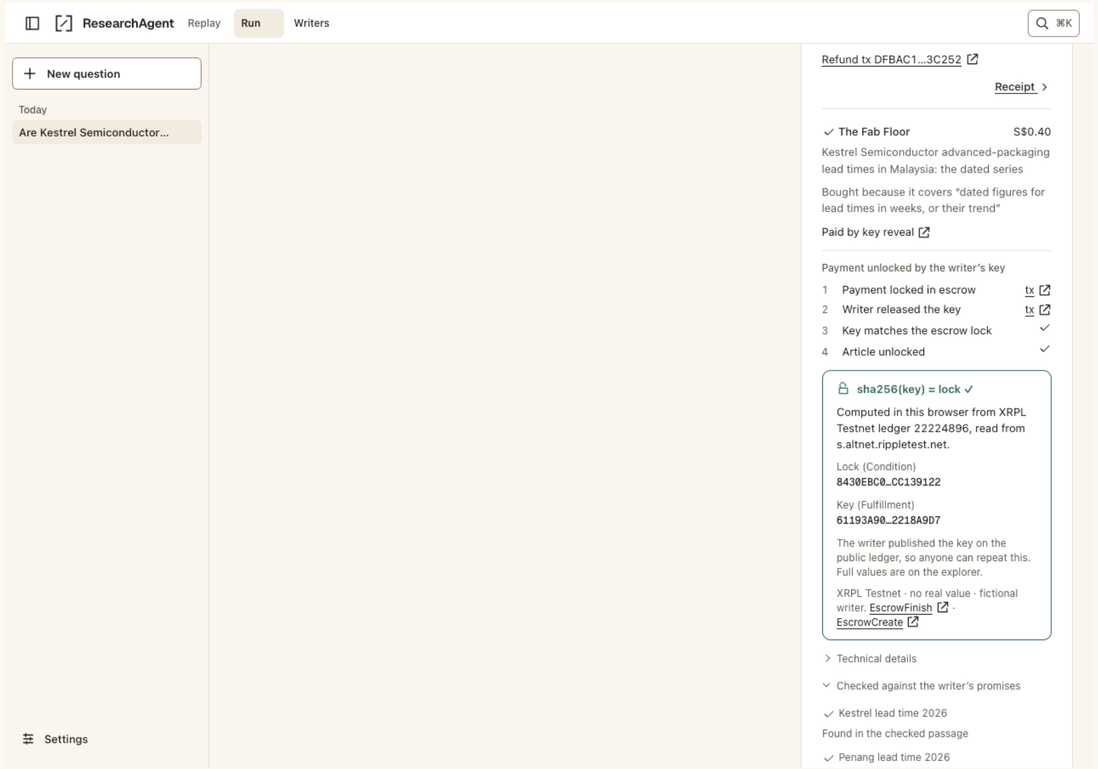
Spending caps for MCP clients default to zero, and the calling model cannot raise them.

— Creativity

Check the delivery key yourself, from the public ledger

- Writer encrypts and signs the price, key hash and content hashes.
- Policy locks payment to the key hash in native Testnet escrow.
- Writer publishes the key in EscrowFinish to collect; the browser checks both transactions.

> A matching key proves this escrow's key reveal, not article quality. Refunds after delivery need the writer's signature; a real writer could refuse. A no-show requires EscrowCancel after CancelAfter.



Local replay UI · recorded Testnet run · fictional writer · public-node check, 11 Nov. The browser recomputes sha256(key) against the lock.

We asked rippled: RLUSD escrow was refused on both networks

TEST-SGD through XLS-85 token escrow

CLI and evidence: the live sandbox pays in Testnet XRP.

- **Opt in** with `--iou` on the MCP server or the web app.
- **Preflight refuses** a missing issuer locking permission, frozen or unauthorized trust lines, short balances and transfer fees.
- **Both verifiers** check currency, issuer and the writer's trust-line balance change, because EscrowFinish has no delivered_amount.

`create E36AF9F1…18878C5C · finish 384CA34F…181C02F5 · anchor 8174E3BE…8ADC35A3`

Unsigned simulate verdicts, 11 Nov

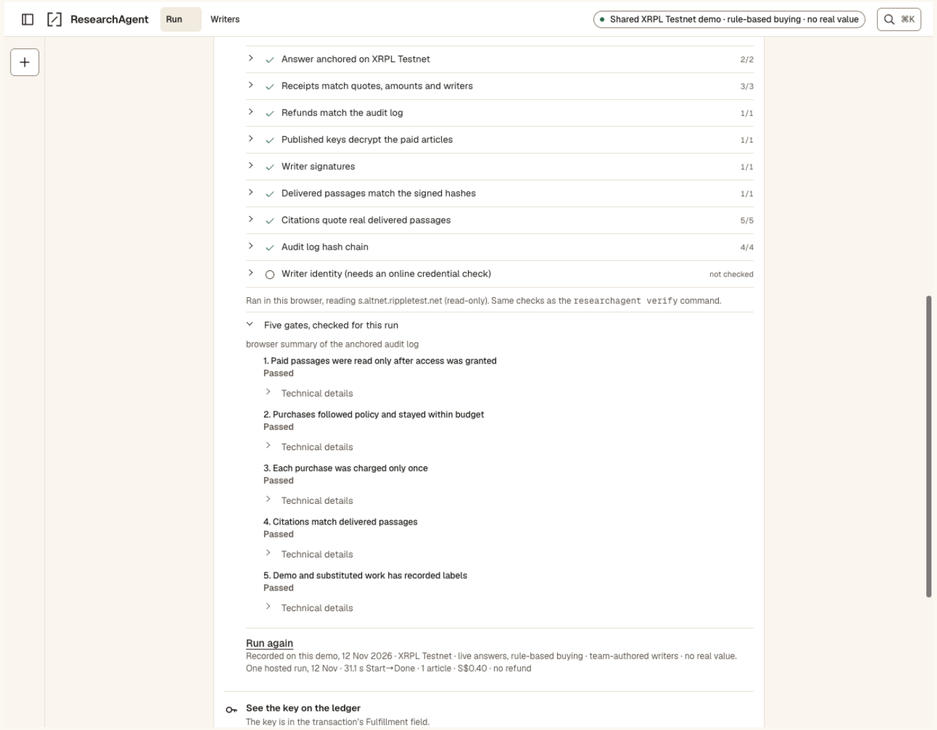
- **Payments work.** Official Testnet RLUSD paid a writer on 3 Nov (`1F2449A4…`), and a broken promise was refunded.
- **Recorded unsigned RPC results.** Nothing signed or submitted.

Create probe	Verdict	Ledger
RLUSD · Testnet	`tecNO_PERMISSION`	`22244249`
RLUSD · Mainnet	`tecNO_PERMISSION`	`108262329`
TEST-SGD · Testnet	`tesSUCCESS`	`22244249`

- **So we escrow our own token** and say so on every amount, receipt and ledger row. TokenEscrow is enabled on Testnet and Mainnet. An issuer change removes this observed blocker; requalification is still needed.

● TEST-SGD · an IOU from our Testnet issuer · stand-in for RLUSD · no peg, backing or mainnet claim · no real value

One hosted run. All five gates held.



Hosted browser, 12 Nov · DEPLOY-5 · recorded preset 2. All five gates Passed (Held); identity skips remain. No question used.

Home → Verify → Run the checks. No question used; public-node reads. Pass, not checked, pending and unavailable stay separate.

- **12 Nov · featured run 421bae3d.** Browser check: all five gates Held. Grant 1953 precedes paid reads 1954–1957.
- **Browser / published TS:** 24 passed, 0 failed, 2 not checked/skipped. **Published Python wheel:** 25 passed, 0 failed, 2 skipped. Testnet 6/6 each.
- **31.1 s Start→Done**, excluding planning/modal; 1 article, S$0.40, no refund. Model cost unmeasured.
- Live answers, rule-based buying; initial fixture draft, keyword fallback, team-authored writer. Testnet, no real value.
- One recorded run. Identity/origin skips remain; this does not prove article quality or every future run.

— Reachability

Every way in runs from one package, for people and for other agents

In the browser

Ask your own question, or check a recorded sandbox run without using a question.

Shared XRPL Testnet demo · live answers, rule-based buying · no real value

Claude Code, any MCP client

`claude mcp add … --testnet`. Spending caps default to zero; the calling model can't raise them.

Cold start 6.3 s · 3.4 MB package

Local app, no clone

`app --testnet` serves the real app on 127.0.0.1 with faucet wallets.

Ready in 27.0 s from an empty cache

TypeScript and Python SDKs

Ask, stream, verify. No purchase method: only backend policy can buy.

The Python verifier needs no Node

Publish your Markdown notes

`publish ./notes --testnet`: faucet wallet, XRPL Credential, key-reveal escrow.

33.9 s from `init` to a validated Credential

Hire the agent over x402

Another agent pays ResearchAgent; it buys, pays every cited writer a royalty and returns the change. Run on Testnet.

One line: `try --testnet --hire`

> **Does it buy the right thing?** v5 blind held-out: 16/20 bought and cited, 0/10 off-topic purchases (run once, 11 Nov, on XRPL Testnet: fixture research and decision models with answers written offline, team-authored writers). The key fact leads 7/20 answers. Packages: GitHub Releases; not yet on npm, PyPI or the MCP Registry.

One paid answer, with its own costs and receipts

How does the writer get paid, and what stops the agent from reading the article and then refusing to pay?

HOSTED-12 · 12 Nov · DEPLOY-3

- **Free draft:** no citable answer; 2 recorded open gaps.
- **Missing:** what binds publishing the article key to payment?
- **Paid p3:** validated EscrowFinish publishes the key and releases payment in the same event.
- **Final answer:** the writer collects by publishing the key; escrow opens only with that key. 0 recorded open gaps; no independent quality score.

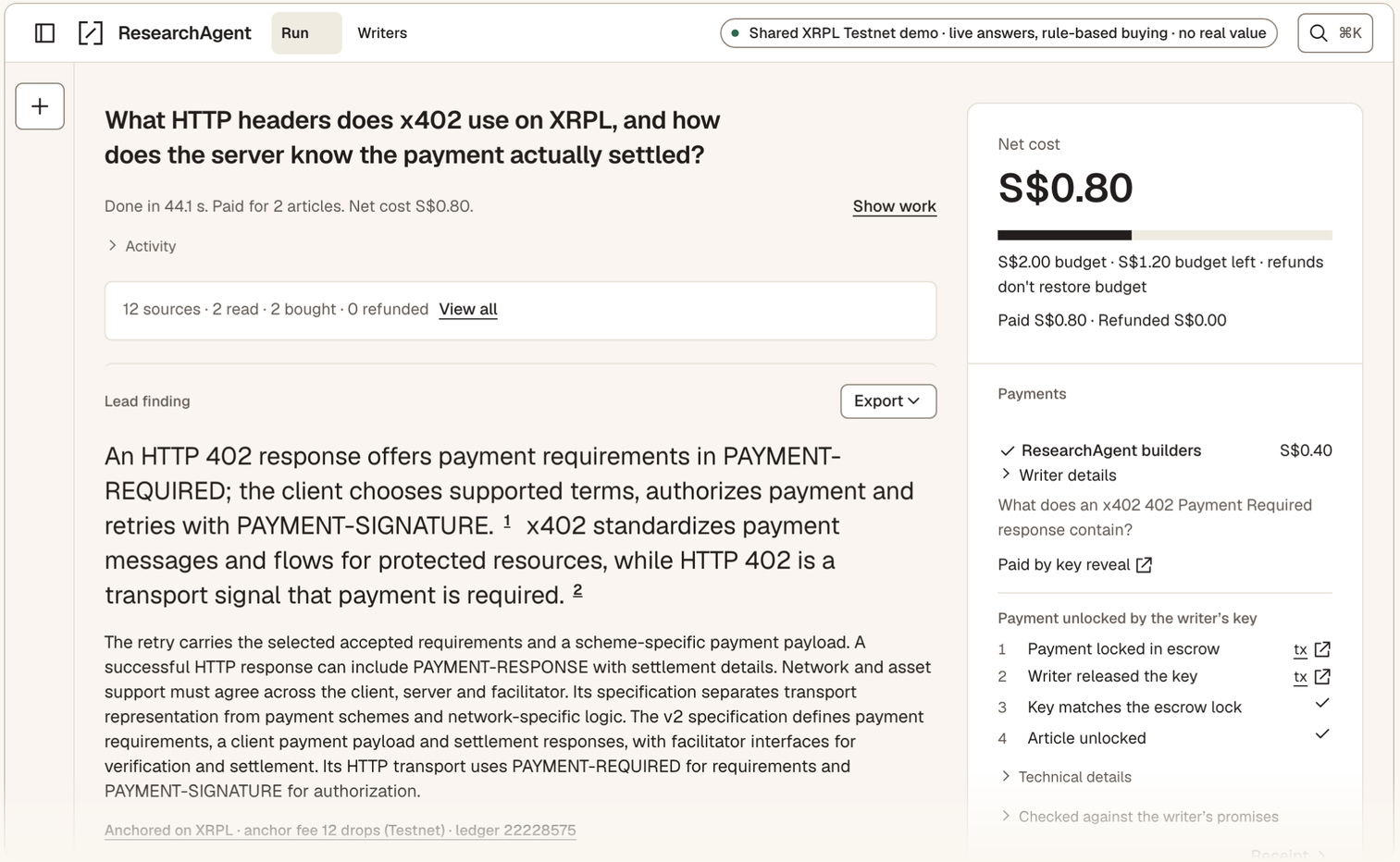
This run’s cost

- 1 article · 4 passages · S$0.40 display price = 40,000 Testnet drops (fixed demo rate, not FX).
- 65.8 s Start→Done; excludes planning and the five-second confirmation modal.
- 12 drops paid by the buyer to create; 423 by the writer to finish; buyer anchor another 12. Model cost: unmeasured (usage not retained).

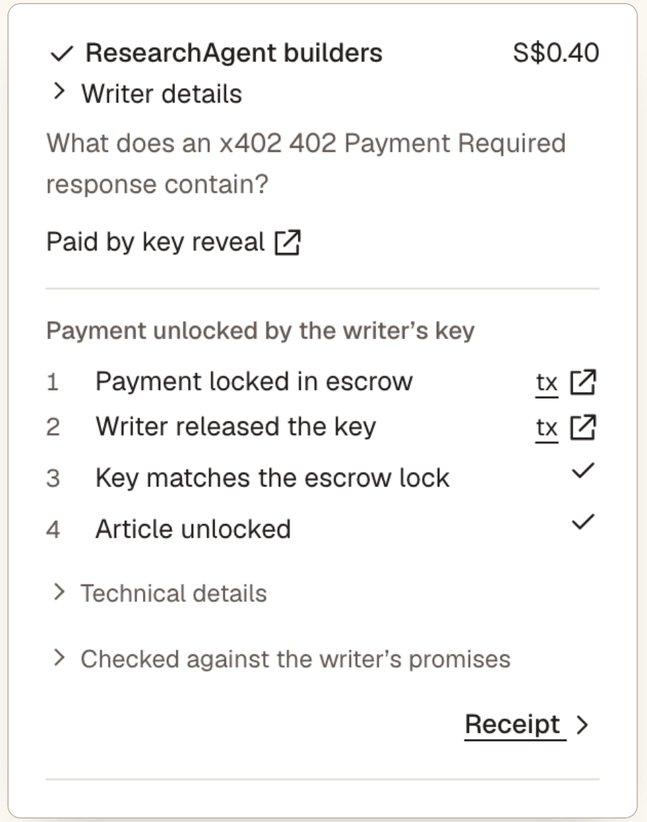
Unsigned rippled autofill: 10 / 350 for the probed shapes. Fees vary with load/key size; article price and anchor are additional.

> One hosted run, 12 Nov · live DeepSeek plan/final answer, rule-based buying · initial draft fixture · keyword fallback · team-authored writer · XRPL Testnet XRP, no real value · no refunds. No general or new held-out claim. Recovery: 0 double charges in 500 synthetic-ledger chaos runs, 130 real process kills.

Recorded answer. Payment and delivery.



11 Nov build · historical hosted answer · actual model label and payment links



Same 11 Nov take: paid by key reveal.

First market hypothesis: XRPL/x402 integrators using coding agents

Buyer and task

An XRPL/x402 integrator asks a coding agent to implement a paid endpoint or diagnose a failed payment.

Seller today

Our team-authored writers sell practitioner implementation notes and reproducible failure paths. Example: how writers get paid.

Expected benefit

Buy the specific implementation detail a free abstract omits, with a cited passage and checkable payment. Test repeat demand here before adding other expert shelves.

> **Initial-market hypothesis, not validated demand.** No independent writers, customers, pilots or revenue. No market sizing. Hire mode charges a service fee on Testnet; its fee and royalty split were demonstrated once.

Our focus: policy evaluates a purchase, checks delivery against signed promises, and can request a writer-signed refund.

Built for audit. Honest about what a regulated launch still needs.

Every spend is authorised by deterministic policy code, capped, and written to a hash-chained audit log whose head can be anchored on the XRP Ledger after a finished answer, even when nothing was bought. It is Testnet only, and each gap has a named next step.

	Built (Testnet)	Not done
Custody	Wallet signing keys stay server-side; article delivery keys become public in EscrowFinish	KMS or HSM and rotation; the hire flow holds funds in transit
Spend limits	A budget per question and a daily cap; only policy code buys	Maker-checker, velocity limits
Audit	Hash-chained log, anchored heads, offline verification	Signed, durable retention; retention periods undecided
Screening, KYB	Payee denylist hook; three accepted operator-issued Testnet writer Credentials; a Testnet Permissioned Domain listing accepted issuer/type for identity display; identity only, not KYC or quality	No sanctions provider, no KYB, no credential-gated payments
Regulation	Questions mapped for MAS (PSA, DPT services), GENIUS Act, Travel Rule	No licence, no legal opinion; mainnet blocked in code

Ranked builder feedback for Ripple, each with a runnable repro

1 **Issuer permission blocks RLUSD escrow**

Recorded unsigned simulate: RLUSD tecNO_PERMISSION on Testnet and Mainnet; TEST-SGD control tesSUCCESS on Testnet. Nothing submitted. Fix: expose ledger verdicts.

2 **An issued amount must mean the delivered amount**

Only validated delivered_amount proves what arrived. Fix: negative issued-asset test vectors.

3 **A signed IOU quote overflowed the 1 KiB Memo**

Observed on Testnet, 7 Nov. Fix: check the serialized Memo size before signing.

4 **Escrow expiry doesn’t return funds by itself**

Funds stayed locked until an explicit EscrowCancel. Fix: document expiry versus cancellation.

5 **Endpoints disagree on retained history**

One answered “not found” for a validated hash; HTTP lacks browser CORS. Fix: an endpoint matrix.

6 **Settled invoice consumption blocks recovery**

A same-payment retry is refused while delivery still needs recovering. Fix: idempotent replay.

7 **A header-only x402 v2 challenge is not retried**

The client returns the unpaid 402. Fix: read the v2 header first.

\+ **Mainnet readiness: a checklist, not a claim**

Plus what worked well: escrow, delivered_amount, Credentials, fast finality.

> 7 ranked findings with runnable repros · 35 numbered findings (B01–B35) · 15 submissions on 4–5 Sep · 8 upstream drafts, 0 filed.

— Team · check every claim

Paid on delivery. Proven on the ledger.

Guruprasath

Product and engineering since SingHacks: plans the work, directs the coding agents and owns the merge gate.

Miul

SingHacks build: the research-plan contracts, quote binding and the first test harness.

Andrew Nguyen

Before SingHacks: the idea briefs and data sources.

How we build

Coding agents work in parallel lanes; every merge passes one local gate of tests and the five hard gates. 304 pull requests since 21 Sep.

Start with “Check our work without asking a question”.

Check it yourself	
Ask, or verify a recorded run	`ra-front-651013803640.asia-southeast1.run.app`
Recorded Testnet run	`collaboration95.github.io/…/demo/`
Ledger page, live checks	`collaboration95.github.io/…/ledger/`
Progress since SingHacks	`collaboration95.github.io/…/progress/`
Release, every artifact	`v1.1.1 on GitHub Releases`
Code and evidence	`github.com/Collaboration95/theFastandtheFungible`
Demo video	Owner video pending (issue #265)

People and roles from the git history (git shortlog) · AGENTS.md (how work is run) · Testnet XRP and TEST-SGD have no value; receipts keep ledger indexes if Testnet resets